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Efficient Synthesis of Functionalized Organozinc Compounds by the Direct Insertion of Zinc into Organic Iodides and Bromides**Arkady Krasovskiy, Vladimir Malakhov,
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Organozinc compounds are important intermediates in organic chemistry because of their compatibility with many functional groups.^[1] The direct insertion of zinc into organic halides is the most attractive and simplest method for the preparation of functionalized organozinc halides.^[1] Whereas several polar solvents such as dimethylacetamide, hexamethylphosphoramide, *N,N*-dimethylformamide, dimethylsulfoxide, acetonitrile, and tetramethylurea can be used,^[2] the most common and convenient is tetrahydrofuran.^[3] The preparation of arylzinc iodides in THF from aryl iodides can only be achieved by using Rieke-zinc^[1b,4] or requires the presence of electron-accepting substituents in the *ortho* position of the aryl iodides as well as elevated reaction temperatures.^[3] The use of Rieke-zinc also allows the preparation of alkylzinc reagents starting from alkyl bromides instead of alkyl iodides.^[5] Since the activity of Rieke-zinc decreases with time, the development of a practical procedure for zinc insertion using commercially available zinc dust would be highly desirable.

Recently, we have found that the Br/Mg exchange reaction can be accelerated by adding LiCl.^[6] Herein, we report a new protocol for the preparation of functionalized aryl- and alkylzinc compounds by the direct insertion of commercially available Zn powder^[7] in the presence of LiCl in THF. Thus, when iodobenzene (**1a**) was treated with commercial zinc powder (3 equiv, 150 mesh, activated by the addition of 1,2-dibromoethane (5 mol %) and TMSCl (1 mol %))^[7] only 5 % conversion could be reached after 24 h at 50 °C. However, in the presence of 1.5 equivalent of LiCl, full conversion into phenylzinc iodide (**1b**) was observed after 7 h at 50 °C. Ketone **1c** was obtained in 90 % yield after reaction with pivaloyl chloride in the presence of CuCN·2 LiCl. The presence of an electron-withdrawing group at the *ortho* (see **2a**) or *para* positions (see **3a**) considerably increases the rate of insertion, and the corresponding arylzinc

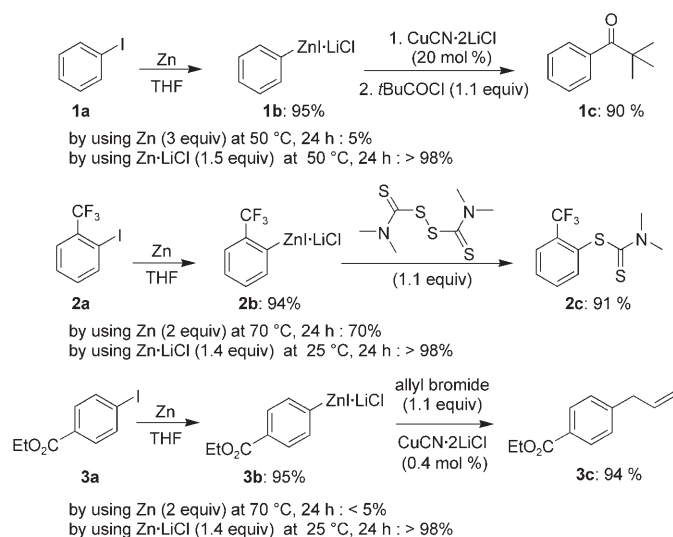
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iodides can be efficiently prepared at room temperature (>98% conversion; Scheme 1). The insertion of zinc in the absence of LiCl can only be accomplished at 70 °C with a low



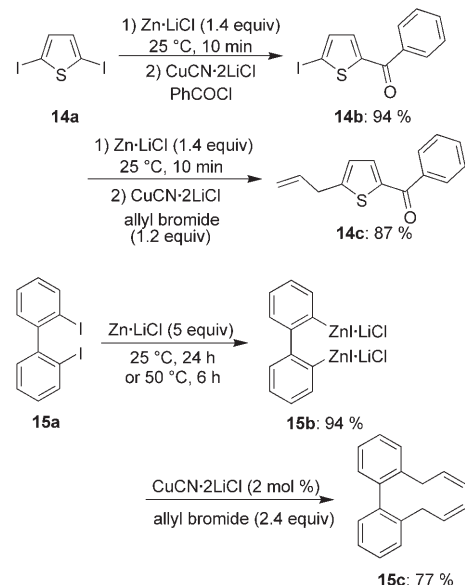
Scheme 1. Insertion of Zn in the presence and absence of LiCl. The specified yields by the formula numbers **1b–3b** were determined by titration; the yields > 98% were determined by GC analysis.

yield for *ortho*-trifluoromethylphenyl iodide (**2a**) and no insertion occurs with ethyl 4-iodobenzoate (**3a**; 70 °C, 24 h, conversion < 5%). The organozinc iodides **2b** and **3b** react cleanly with electrophiles such as tetramethylthiuram disulfide^[8] and allyl bromide to afford the corresponding products **2c** and **3c** in yields of 91 and 94%, respectively (Scheme 1).

4-Anisylzinc iodide (**4a**) can be prepared in 93% yield by the insertion of Zn in the presence of LiCl at 50 °C in THF (entry 1, Table 1).^[9] A broad range of functionalized arylzinc iodides bearing an ester, a nitrile, or an amide group such as **5a–8a** can be easily prepared in good yields (92–95%) and treated with electrophiles (entries 2–5, Table 1). A sensitive methyl ketone functionality is also compatible with the mild insertion conditions. 4-Acetylphenylzinc iodide (**9a**) was formed in 87% yield after 3 h at 25 °C and a copper-catalyzed acylation led to the diketone **9b** in 88% yield (entry 6, Table 1). Interestingly, even an aldehyde group can be tolerated. Thus, aryl- and heteroarylzinc iodides bearing a formyl group (**10a** and **11a**) were obtained in good yields by a direct zinc insertion reaction at room temperature (entries 7 and 8, Table 1). Transition-metal-catalyzed coupling reactions with ethyl 4-iodobenzoate (1 mol % [Pd(PPh₃)₄]) and ethyl 2-bromomethyl acrylate (0.4 mol % of CuCN·2LiCl) led to the desired functionalized aldehydes **10b** and **11b** in yields of 85 and 81%, respectively (entries 7 and 8, Table 1).

The insertion reaction displays an excellent chemoselectivity. Thus, treating 4-bromophenyl iodide (50 °C, 6 h) with Zn/LiCl resulted only in insertion into the C–I bond to give the zinc reagent **12a** (96%). Subsequent allylation provides the aryl bromide **12b** in 96% yield (entry 9, Table 1). Moreover, a selective formation of the corresponding mono-metalated compound is possible in the case of diiodoaryl

compounds (**13a**, entry 10, Table 1). A stepwise insertion into 2,5-diiodothiophene (**14a**) allows the introduction of two different electrophiles (Scheme 2). Thus, the reaction of diiodide **14a** with Zn/LiCl followed by a copper(I)-catalyzed



Scheme 2. Chemoselective zinc insertion into diiodides.

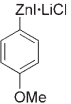
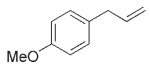
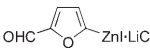
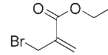
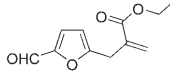
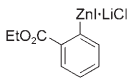
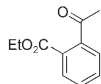
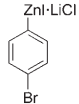
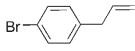
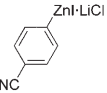
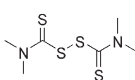
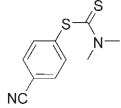
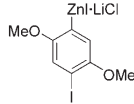
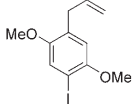
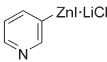
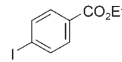
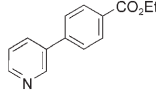
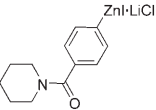
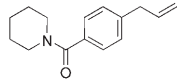
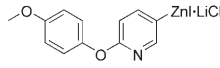
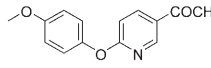
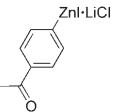
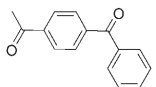
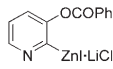
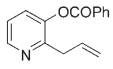
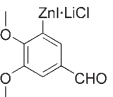
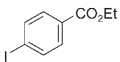
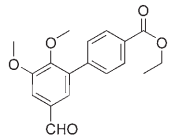
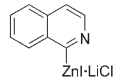
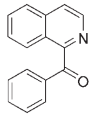
acylation led to the ketone **14b**, which can be further functionalized through a zinc insertion/allylation sequence to afford the disubstituted thiophene **14c** in 82% overall yield. Alternatively, the formation of a bismetallic species can be achieved.^[10] Thus, 2,2'-diiodobiphenyl (**15a**) reacts smoothly with Zn/LiCl (5 equiv) at 50 °C to furnish bis(iodozincio)biphenyl (**15b**) in 94% yield. The corresponding biphenyl derivative **15c** was obtained in 77% yield after a copper(I)-catalyzed coupling with allyl bromide.

A smooth zinc insertion is also observed for heterocyclic iodides. Thus, various iodopyridines and 1-iodoisoquinoline react readily with Zn powder to afford the corresponding heteroarylzinc iodides **16a–19a** at RT in yields of greater than 90% (entries 11–14, Table 1). The resulting organozinc compounds were efficiently quenched with a wide range of electrophiles under transition-metal catalysis to give the functionalized aza-heterocycles **16b–19b** in yields of 83–90%.

The insertion of Zn into a C–Br bond is also possible in the case of activated aryl- and heteroaryl compounds. Thus, in the presence of LiCl, the functionalized zinc reagent **20b** was formed at room temperature in 91% yield starting from the bromodiester **20a**. Quenching with allyl bromide leads to the allylated diester **20c** in 90% yield. Similarly, the reaction of ethyl 5-bromo-2-furoate (**21a**) with Zn/LiCl furnishes the corresponding furylzinc reagent **21b** in 92% yield after 12 h at 25 °C. This compound was successfully coupled with 4-iodobenzonitrile in the presence of [Pd(PPh₃)₄] (1 mol %; Scheme 3).^[11]

For the first time, the use of Zn powder in the presence of LiCl allows the expedient preparation of functionalized

Table 1: Preparation and reactions of functionalized aryl- and heteroarylzinc reagents.

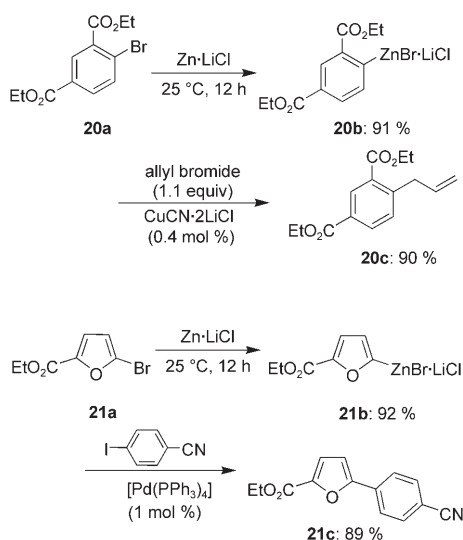
Entry	Zinc reagent, yield [%] ^[a]	T [°C], t [h]	Electrophile	Product, yield [%] ^[b]	Entry	Zinc reagent, yield [%] ^[a]	T [°C], t [h]	Electrophile	Product, yield [%] ^[b]
1	 4a : 93	50, 90	AllBr ^[c]	 4b : 94	8	 11a : 83	25, 2.5	 Br-CH=CH-COOEt	 11b : 81
2	 5a : 92	25, 1	AcCl ^[d]	 5b : 90	9	 12a : 96	50, 6	AllBr ^[c]	 12b : 96
3	 6a : 95	50, 3		 6b : 89	10	 13a : 90	50, 120	AllBr ^[c]	 13b : 89
4	 7a : 93	50, 6	PhCOCl ^[d]	 7b : 88	11 ^[e]	 16a : 92	25, 8	 I-C ₆ H ₄ -CO ₂ Et	 16b : 87
5	 8a : 93	25, 24	AllBr ^[c]	 8b : 96	12	 17a : 90	25, 24	CH ₃ COCl ^[d]	 17b : 83
6	 9a : 87	25, 3	PhCOCl ^[d]	 9b : 88	13	 18a : 91	25, 12	AllBr ^[c]	 18b : 85
7 ^[e]	 10a : 82	25, 12	 I-C ₆ H ₄ -CO ₂ Et	 10b : 85	14	 19a : 96	25, 12	PhCOCl ^[d]	 19b : 90

[a] Yield estimated after titration with I_2 . [b] Yield of isolated analytically pure product based on the molarity of the zinc reagent. [c] 2 mol% of $CuCN \cdot 2 LiCl$ was added. [d] 20 mol% of $CuCN \cdot 2 LiCl$ was added. [e] 1 mol% of $[Pd(PPh_3)_4]$ was added.

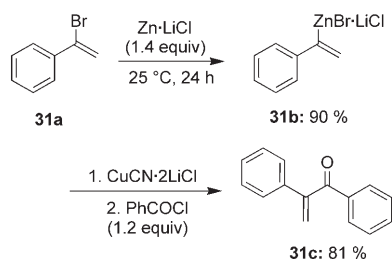
alkylzinc reagents **22a–30a** starting from alkyl bromides (Table 2). Thus, unactivated *n*-octylbromide and 1-bromo-5-chloropentane react cleanly with Zn-LiCl to afford the corresponding alkylzinc reagents in yields of greater than 92 %. The reaction with PhCOCl in the presence of catalytic amounts of Pd⁰ provides ketones **22b** and **23b** in 89 % yield (Table 2).^[11] The formation of cyclobutylmethylzinc bromide **24a** proceeds without the formation of ring-opening products, thus leading after allylation to the unsaturated ester **24b**. Functionalized alkylzinc compounds such as **25a–27a** can also be prepared in high yields. The Pd-catalyzed coupling of these zinc reagents with different ethyl iodobenzoates lead to difunctionalized benzenes **25b–27b** in good yields (entries 4–6, Table 2). The insertion reaction is also efficient in the case

of secondary alkyl bromides and provides alkylzinc reagents **28a** and **29a** in yields of 96 and 93 %, respectively. Adamantylzinc bromide (**30a**) is obtained in 83 % yield by the direct insertion of Zn in the presence of LiCl. Secondary alkylzinc reagents **28a–30a** show good reactivity towards electrophiles (entries 7–9, Table 2). Finally, zinc insertion in the presence of LiCl allows the smooth preparation of the alkenylzinc reagent **31b** starting from α -bromostyrene (**31a**). Copper(I)-catalyzed acylation with benzoyl chloride leads to the unsaturated ketone **31c** in 81 % yield (Scheme 4).

The nature of the activation of the zinc dust with LiCl is speculative. However, it can be envisioned that the role of LiCl is to rapidly remove the formed organozinc reagent from the metal surface by generating highly soluble $\text{RZnX} \cdot \text{LiCl}$.



Scheme 3. LiCl-mediated zinc insertion into aryl and heteroaryl bromides.



Scheme 4. Insertion of Zn into α -bromostyrene.

complexes, thus allowing a rapid reaction of a further molecule of R–X and thereby avoiding the competitive deactivation of the active metal sites.

In conclusion, we have shown that the use of Zn powder in the presence of LiCl in THF allows a simple, high-yielding preparation of a broad range of functionalized aryl- and heteroarylzinc reagents. The synthesis of the alkylzinc reagents was performed from inexpensive and readily available alkyl bromides. All reactions proceed within a practical temperature range (25–50 °C), and can be extended to large-scale preparation. Extensions of this work are currently underway.^[12]

Experimental Section

Typical procedure. Preparation of **3c** (Scheme 1): Anhydrous LiCl (5 mmol) was placed in an Ar-flushed flask and dried for 20 min at 150–170 °C under high vacuum (1 mbar). Zinc powder (7 mmol, 1.4 equiv, 150 mesh, Chemetall, or 325 mesh, Strem) was added under Ar and the heterogeneous mixture of Zn and LiCl was dried again for 10–20 min at 150–170 °C under high vacuum (1 mbar). The reaction flask was evacuated and refilled with argon three times. THF (5 mL) was added and the Zn was activated with BrCH₂CH₂Br (5 mol %) and Me₃SiCl (1 mol %).^[7] **3a** (5 mmol) was then added neat at room temperature. The insertion reaction was complete after 24 h (checked by GC analysis of reaction aliquots, the conversion was higher than 98 %). Titration of an aliquot (1 mL) of the organozinc reagent with

Table 2: Preparation and reaction of aliphatic zinc reagents.

Entry	Zinc reagent, yield [%] ^[a]	T [°C] t [h]	Electrophile	Product, yield [%] ^[b]
1	<i>n</i> -C ₈ H ₁₇ ZnBr·LiCl 22a : 92	50, 24	PhCOCl ^[c]	<i>n</i> -Octyl phenyl ketone 22b : 89
2	Cl(CH ₂) ₅ ZnBr·LiCl 23a : 93	50, 12	PhCOCl ^[c]	Cl(CH ₂) ₅ phenyl ketone 23b : 89
3	Cyclopropylmethyl ZnBr·LiCl 24a : 86	50, 50	BrCH ₂ CO ₂ CH ₃ ^[d]	Cyclopropylmethyl 3-oxopropyl ether 24b : 82
4	AcO(CH ₂) ₄ ZnBr·LiCl 25a : 96	50, 3	I-Ph-CO ₂ Et	AcO(CH ₂) ₄ -Ph-CO ₂ Et 25b : 83
5	EtO ₂ C(CH ₂) ₃ ZnBr·LiCl 26a : 95	50, 1	I-Ph-CO ₂ Et	EtO ₂ C(CH ₂) ₃ -Ph-CO ₂ Et 26b : 93
6	NC(CH ₂) ₄ ZnBr·LiCl 27a : 95	50, 6	I-Ph-CO ₂ Et	NC(CH ₂) ₄ -Ph-CO ₂ Et 27b : 88
7	Cyclohexyl ZnBr·LiCl 28a : 96	50, 24	PhCOCl ^[c]	Cyclohexyl phenyl ketone 28b : 94
8	1-Hexyl ZnBr·LiCl 29a : 93	50, 18	PhCOCl ^[c]	1-Hexyl phenyl ketone 29b : 90
9	Bicyclo[2.2.1]hept-5-en-2-yl ZnBr·LiCl 30a : 83	50, 24	(CH ₃) ₂ N-C(=S)-S-C(=S)-N(CH ₃) ₂	Bicyclo[2.2.1]hept-5-en-2-yl thioacetate 30b : 79

[a] Yield determined by titration with I₂. [b] Yield of isolated analytically pure product based on the molarity of the zinc reagent. [c] 0.1 mol % of [Pd(dba)₂] (dba = *trans,trans*-dibenzylideneacetone) and 0.2 mol % of *S*-phos^[14] were added. [d] 1 mol % of CuCN·2LiCl was added.

iodine^[13] showed that the concentration of **3b** in THF was 0.83 M. The solution of **3b** (3 mmol, 3.61 mL) was carefully separated from the remaining zinc powder by using a syringe and transferred to another dry and Ar-flushed flask. Allyl bromide (3.3 mmol) was added at –20 °C, followed by CuCN·2LiCl (one drop of a 1.0 M solution in THF, ca. 0.02 mmol, ca. 0.4 mol %). The reaction mixture was stirred for 1 h at 0 °C and was then quenched with saturated aqueous NH₄Cl solution (5 mL). The aqueous phase was extracted with ethyl acetate (3 × 5 mL) and concentrated in vacuo. The crude residue was purified by flash chromatography (CH₂Cl₂) and yielded **3c** (2.82 mmol, 504 mg, 94 %) as a colorless oil.

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- [1] a) "Polyfunctional Zinc Organometallics for Organic Synthesis": P. Knochel, H. Leuser, L.-Z. Gong, S. Perrone, F. F. Kneisel, *Handbook of Functionalized Organometallics, Vol. 1* (Ed.: P. Knochel), Wiley-VCH, Weinheim, **2005**, p. 251; b) *Organozinc Reagents* (Eds.: P. Knochel, P. Jones), Oxford University Press, New York, **1999**.
- [2] a) K. Tagaki, N. Hayama, S. Inokawa, *Bull. Chem. Soc. Jpn.* **1980**, *53*, 3691; b) K. Tagaki, *Chem. Lett.* **1993**, 469; c) K. Tagaki, Y. Shimoishi, K. Sasaki, *Chem. Lett.* **1994**, 2055; d) T. N. Majid, P. Knochel, *Tetrahedron Lett.* **1990**, *31*, 4413; e) C. Gosmini, M. Amatore, S. Claudel, J. Perichon, *Synlett* **2005**, 2171; f) I. Kazmierski, C. Gosmini, J.-M. Paris, J. Perichon, *Tetrahedron Lett.* **2003**, *44*, 6417.
- [3] R. Ikegami, A. Koresawa, T. Shibata, K. Takagi, *J. Org. Chem.* **2003**, *68*, 2195.
- [4] a) R. D. Rieke, P. T. Li, T. P. Burns, S. T. Uhm, *J. Org. Chem.* **1981**, *46*, 4323; b) R. T. Arnold, S. T. Kulenovic, *Synth. Commun.* **1977**, *7*, 223.
- [5] a) L. Zhu, R. M. Wehmeyer, R. D. Rieke, *J. Org. Chem.* **1991**, *56*, 1445; b) R. D. Rieke, M. V. Hanson, J. D. Brown, Q. J. Niu, *J. Org. Chem.* **1996**, *61*, 2726; c) P. Knochel, M. C. P. Yeh, S. C. Berk, J. Talbert, *J. Org. Chem.* **1988**, *53*, 2390.
- [6] a) A. Krasovskiy, P. Knochel, *Angew. Chem.* **2004**, *116*, 3396; *Angew. Chem. Int. Ed.* **2004**, *43*, 3333; b) H. Ren, A. Krasovskiy, P. Knochel, *Chem. Commun.* **2005**, 543; c) H. Ren, A. Krasovskiy, P. Knochel, *Org. Lett.* **2004**, *6*, 4215; d) A. Krasovskiy, B. F. Straub, P. Knochel, *Angew. Chem.* **2006**, *118*, 165; *Angew. Chem. Int. Ed.* **2006**, *45*, 159.
- [7] The zinc powder was activated by treatment first with 1,2-dibromoethane and then with chlorotrimethylsilane; see: P. Knochel, M. J. Rozema, C. E. Tucker, *Organocopper reagents* (Ed.: R. J. K. Taylor), Oxford University press, Oxford, **1994**, p. 85.
- [8] A. Krasovskiy, A. Gavryushin, P. Knochel, *Synlett* **2006**, 792.
- [9] There is only one literature precedent^[3] for the direct synthesis of **4a** in an ethereal solvent (24 h, 130 °C in diglyme).
- [10] a) J. F. Normant, I. Marek, J. M. Lefrancois, *Pure Appl. Chem.* **1992**, *64*, 1857; b) J. F. Normant, *Acc. Chem. Res.* **2001**, *34*, 640.
- [11] "Palladium- or nickel-catalyzed cross-coupling with organometals containing zinc, aluminum, and zirconium: The Negishi coupling": a) E. Negishi, X. Zeng, Z. Tan, M. Qian, Q. Hu, Z. Huang, *Metal-Catalyzed Cross-Coupling Reactions, Vol. 2, 2nd ed.* (Eds.: A. De Meijere, F. Diederich), Wiley-VCH, Weinheim, **2004**, p. 815.
- [12] A patent application was filed.
- [13] A. Krasovskiy, P. Knochel, *Synthesis* **2006**, 890.
- [14] S. D. Walker, T. E. Barder, J. R. Martinelli, S. L. Buchwald, *Angew. Chem.* **2004**, *116*, 1907; *Angew. Chem. Int. Ed.* **2004**, *43*, 1871.